

Question_4:

Code:

```
1  /*
2  QUESTION_4:
3  Given a linked list, insert a new node before a given node P.
4  */
5  #include <iostream>
6  using namespace std;
7
8  struct node {
9      int data;
10     node *next;
11 };
12 void display(node *start) {
13     cout << "Linked List:";
14     while (start) {
15         cout << "->" << start->data;
16         start = start->next;
17     }
18 }
19 int main() {
20     int n, x, p;
21     cin >> n;
22
23     node *head = NULL;
24     node *tail = NULL;
25
26     for (int i = 0; i < n; i++) {
27         cin >> x;
28         node *t = new node{x, NULL};
29         if (!head)
30             head = tail = t;
31         else {
32             tail->next = t;
33             tail = t;
34         }
35     }
36     cin >> p;
37     cin >> x;
38 }
```

```
39     node *newNode = new node{x, NULL};
40
41     if (head && head->data == p) {
42         newNode->next = head;
43         head = newNode;
44         display(head);
45         return 0;
46     }
47     node *t = head;
48
49     while (t && t->next && t->next->data != p)
50         t = t->next;
51     if (!t || !t->next) {
52         cout << "Node not found!";
53         display(head);
54     }
55     else {
56         newNode->next = t->next;
57         t->next = newNode;
58         display(head);
59     }
60     return 0;
61 }
```

Output:

Test Case_1:

```
4
115 210 116 110
116
17
Linked List:->115->210->17->116->110
PS C:\Users\Prabin Yadav\Documents\III
```

Test Case_2:

```
4
15 20 16 10
25
17
Node not found!Linked List:->15->20->16->10
PS C:\Users\Prabin Yadav\Documents\III SEM\DS
```